CuteMagic SkyIsland Pack Free 01

Documentation

Welcome to the **SkyIsland Pack Free 01** pack!

This documentation is designed to help you quickly integrate the assets into your Unity projects, understand the package structure, and make the most of its technical features and performance optimizations.

1. **Introduction**

   Thank you for choosing **SkyIsland Pack Free 01**!

   This asset pack is built around three core principles: **production-ready quality**, **stylized low-poly design**, and **optimized real-time performance**.

   Every asset features clean topology, a consistent visual style, and an optimization-focused workflow, making it suitable for projects ranging from mobile games to PC, console, and VR/AR applications.

   From clean geometry and lightweight shaders to shared materials and texture atlases, every technical decision has been made to reduce rendering overhead while maintaining professional production standards.

2. **Package Contents**
   The package includes:

   - **Prefabs**

     Ready-to-use Unity prefabs with materials already configured.

     Simply drag and drop them into your scene.

   - **Source Models**

     Original FBX models for customization or editing.

   - **Materials**

     Shared materials configured for the supported render pipeline.

   - **Textures**

     Optimized texture atlas and texture resources used throughout the package.

- **Demo Scene**

  A demonstration scene showing the recommended setup and example usage.

3. **Performance Optimization**

   Performance has been a primary focus throughout the development of this asset pack.

   Optimized Geometry

   Every model features clean, production-ready topology with low vertex and polygon counts for efficient real-time rendering.

   Benefits:

   - Lower GPU workload
   - Faster rendering
   - Better scalability
   - Suitable for large scenes

   Texture Atlas

   All assets share a single palette-based texture atlas.

   Benefits:

   - Reduced texture switching
   - Lower memory usage
   - Improved batching
   - Fewer draw calls

   Shared Materials

   All assets share a single master material.

   Benefits:

   - Reduced draw calls
   - Improved batching
   - Easier material management

   Lightweight Shaders

Simple, optimized shaders help reduce CPU and GPU overhead while maintaining the intended visual style.

4. **Render Pipeline Support**
   This package is fully compatible with:

   ✔ Universal Render Pipeline (URP)

   All materials and shaders are preconfigured and ready to use immediately after importing the package.

   Recommended Unity Version:

   2022.3 LTS or newer

5. **Shader Optimization**

   The shaders included in this package are specifically designed for efficient real-time rendering while maintaining a clean, stylized appearance.

   - **Lightweight PBR Workflow**

     A streamlined Physically Based Rendering (PBR) implementation delivers high visual quality with minimal rendering overhead, making it suitable for mobile, PC, console, and VR/AR projects.

   - **Channel-Packed PBR Textures**

     To reduce memory usage and texture sampling costs, PBR surface data is packed into a single texture:

     - ❖ R: Metallic
     - ❖ G: Ambient Occlusion
     - ❖ B: Roughness

     This approach minimizes texture fetches while preserving material quality.

   - **Shared Material Workflow**

     The shader is designed to work with a shared material and texture atlas workflow, helping reduce draw calls and improve batching efficiency across the entire asset pack.

   - **Material Customization**

Common surface properties - including Base Color Tint, Metallic, Roughness, Ambient Occlusion, and Normal Strength - can be adjusted directly in the Material Inspector without modifying the source textures.

- **URP Native Implementation**

  Built directly on Unity's Universal Render Pipeline (URP), the shader integrates seamlessly with Unity's standard lighting system while remaining lightweight and easy to maintain.

- **GPU Instancing Support**

  Supports Unity GPU Instancing for compatible rendering scenarios.

6. How to Use

- **Importing the Package**
  Import the asset package into a Unity project using the supported Render Pipeline specified in this documentation.

  Once imported, all assets, materials, and textures are ready to use without additional setup.

- **Adding Assets to Your Scene**
  Open the Prefabs folder in the Project window.

  Drag and drop any prefab into your Scene or Hierarchy.

  Position, rotate, and scale the assets as needed to build your environment.

- **Customizing Materials**
  The package is designed around a shared material workflow to maximize rendering performance.

  You can customize the shared material properties directly in the Material Inspector. Depending on the asset pack, available parameters may include color tint, surface properties, normal strength, or other shader-specific controls.

  Changes made to the shared material are automatically applied to all assets using that material.

- **Modifying Source Models (Optional)**
  Original FBX source models are included for further customization.

If required, you can edit the models using your preferred 3D software (such as Blender, Maya, or 3ds Max) and re-import them into Unity while preserving the existing material workflow.

7. **Folder Structure**

   SkyIsland Pack Free 01

```
├── Prefabs
├── Models (FBX)
├── Materials
│   ├── Master_Material
├── Textures
│   ├── Master_Texture.png
├── Shaders
│   ├── Mobile_HighPerformance_PBR.shader
├── Scenes
└── Documentation
```

8. **Technical Specifications**
   - Prefabs: 300+.
   - Source Models: 300+ FBX.
   - Average Geometry: ~300 vertices per model.
   - Materials: 1 Shared Master Material.
   - Texture Setup: 1 Palette-Based Texture Atlas.
   - Texture Resolution: 2048 x 2048 (PNG).
   - Shader: Mobile High-Performance PBR.
   - Draw Call Optimization: Single Material + Texture Atlas Workflow.
   - Supported Render Pipeline: Universal Render Pipeline (URP).
   - Recommended Unity Version: 2022.3 or newer.
9. **Support & Updates**

   Future updates, improvements, and content additions will be provided free of charge for existing customers.

If you have questions, encounter any issues, or would like to suggest new features or future asset packs, we would love to hear from you.

Contact

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Thank You

Thank you for supporting **SkyIsland Pack Free 01**.

We hope this package helps accelerate your workflow and contributes to creating beautiful, optimized projects.

Your feedback is always appreciated and helps us continue improving future releases.